Marked primitive components and the expansion conjecture

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Task 1. Full solution

For $a, k \geq 1$, let $\mathcal{S}_{n,k}^{a\prec n}(1324)$ consist of the permutations of $[n]$ avoiding 1324 in which a is k positions to the left of n, and all letters $1, \ldots, a-1$ are to the right of n. Write

$$T_{a,k}(x) = \sum_{n} |\mathcal{S}_{n,k}^{a\prec n}(1324)| x^n, \qquad T_{a,0}(x) = |\operatorname{Av}_{a-1}(1324)| x^a.$$

The auxiliary class at $k = 0$ consists of the words $a\lambda$, where λ is a 1324-avoiding permutation of $[a-1]$. The empty permutation is allowed, so $T_{1,0} = x$.

A *primitive* is a 1324-avoiding permutation whose minimum is immediately followed by its maximum. If $\mathcal{P}_m$ is this class in size $m \geq 2$, the series used in the source is

$$f(x) = \sum_{m \geq 2} |\mathcal{P}_m| x^{m-1} = x + 2x^2 + 6x^3 + 22x^4 + 91x^5 + \cdots .$$

We use the primitive factorization of Gil–Lopez–Weiner [1, Propositions 2.3–2.4 and Corollary 2.5]. Their paper proves the cases $a = 1, 2$ and also gives the marking argument for $a = 3$. The following proof extends that method to every a by proving that unmarked components can always be inserted.

Theorem 1. *For each $a \geq 1$ there are series $H_{a,r}(x)$ with nonnegative integer coefficients, zero for $r \geq a$, such that for every $k \geq 0$,*

$$T_{a,k} = \sum_{r=0}^{\min(k,a-1)} \binom{k}{r} f^{k-r} H_{a,r}. \tag{1}$$

Consequently, for every $k \geq a$ all three formulas of [1, Conjecture 1] hold:

$$\sum_{j=0}^{k} (-1)^j \binom{k}{j} f^j T_{a,k-j} = 0, \tag{2}$$

$$T_{a,k} = \sum_{r=0}^{a-1} \binom{k}{r} f^{k-r} \sum_{i=0}^{r} (-1)^i \binom{r}{i} f^i T_{a,r-i}, \tag{3}$$

$$T_{a,k} = \sum_{j=0}^{a-1} (-1)^{a-j-1} \binom{k}{j} \binom{k-j-1}{a-j-1} f^{k-j} T_{a,j}. \tag{4}$$

All identities are identities of formal power series in x.

The primitive components. Call letters below a *small*, and letters at least a *large*. Delete the small letters from an object counted by $T_{a,k}$ and standardize. The cited unique factorization gives k primitives. Before standardizing, this large-letter word has the form

$$A_k \cdots A_1 \; a \, b_1 \cdots b_k \; B_k \cdots B_1, \qquad a = b_0 < b_1 < \cdots < b_k = n. \tag{5}$$

For each i, the words A_i, B_i partition the interval of integer letters (b_{i-1}, b_i). The word $A_i \, b_{i-1} \, b_i \, B_i$ standardizes to the ith primitive. Conversely, any ordered list of primitives gives a unique avoiding word of the form (5). In particular, every large letter in B_i exceeds every large letter in B_j when $i > j$. The suffix $B_k \cdots B_1$ avoids 213: otherwise prepending the earlier letter a would give 1324.

A canonical assignment of the small letters. In the suffix to the right of n, attach each maximal run of small letters that is followed by a large letter to that following letter. The final run, if any, is kept separately as a tail L. Let D_i be B_i with its attached runs inserted. Thus the full permutation has the unique form

$$A_k \cdots A_1 \; a \, b_1 \cdots b_k \; D_k \cdots D_1 \; L. \tag{6}$$

Every nonempty D_i ends in a large letter, and D_i is empty if B_i is empty. Mark component i precisely when D_i contains a small letter. The marked components use disjoint nonempty sets of the $a-1$ small letters; hence their number r is at most $a-1$. An unmarked component has $D_i = B_i$.

Lemma 1 (Deletion and arbitrary reinsertion)**.** *Delete all unmarked components in* (6)*, meaning that for each such i we delete A_i, B_i, and the upper endpoint b_i. Standardize the result, leaving the small letters $1, \ldots, a-1$ and the base letter a fixed. The result is an avoiding permutation with exactly r components, all marked, or an auxiliary $k = 0$ object if $r = 0$.*

Conversely, into any such fully marked object one may insert arbitrary unmarked primitives at arbitrary places in its ordered component list. This always produces a 1324*-avoiding permutation.*

Proof. Deletion produces a subsequence followed by an order-preserving relabeling, so it preserves avoidance. Each retained component keeps its internal order and decorations. Its width $b_i - b_{i-1}$ is unchanged; the successive intervals are simply made consecutive again. The unique primitive factorization of the remaining large-letter word is therefore exactly the retained list. Each small run still has its assigned following large letter, and the tail stays at the end. This proves the deletion assertion, including $r = 0$, when only aL remains.

For the converse, form (6) with the extra primitives, using consecutive intervals for their large letters and preserving all small letters and their assignments. Its large-letter subsequence avoids 1324 by the cited primitive-product theorem. Its small-letter subsequence is unchanged. Suppose nevertheless that a forbidden pattern appears. Classify it by its number of small letters.

With zero small letters it contradicts avoidance of the large-letter subsequence. With four small letters it was already in the core. With at least one small letter the first letter of the pattern must be small: the first letter is the minimum of a 1324 pattern. Thus the entire pattern lies to the right of the new maximum, in $D_k \cdots D_1 L$.

With exactly one small letter, the three large letters would form 213 in $B_k \cdots B_1$, which is impossible.

With exactly two small letters, the pattern has the form s_1, u, s_2, v, where $s_1 < s_2 < a \le u < v$. An increasing pair u, v cannot cross two suffix blocks, since those blocks occur in decreasing order of their value intervals. Hence u, v lie in the same B_i. This cannot be an inserted unmarked block, because s_2 lies between u and v. It is therefore a retained marked block, and the same four letters form a forbidden pattern in the core.

With exactly three small letters, those letters form 132 and are followed by a large letter v. If v is retained, this pattern is already in the core. If v belongs to an inserted block, consider the last of the three selected small letters, in positional order. It cannot lie in the final tail. It lies in a retained decorated block D_i before the inserted block. Its attached run is followed by a retained large letter w in that same D_i, hence after all three selected small letters and before v. Replacing v by w gives 1324 in the core, again a contradiction. These cases exhaust all possibilities. □

Proof of Theorem 1. Let $H_{a,r}$ count, by their full size, the avoiding objects with r components all marked, retaining every small letter and the tail. For $r = 0$ this is the auxiliary class, so $H_{a,0} = T_{a,0}$. The marking bound gives $H_{a,r} = 0$ for $r \geq a$.

Lemma 1 is a bijection: an object with k components determines its core, the r marked positions in the component list, and its ordered $k - r$ unmarked primitives. Conversely these data recover exactly one object. No labels need to be chosen: consecutive large-letter intervals determine them, while the small labels belong to the core. A primitive of size m adds $m - 1$ letters, since its lower endpoint is shared with the preceding part of the increasing central sequence. Thus every unmarked primitive contributes f, and the choices of marked positions contribute $\binom{k}{r}$. This proves (1).

Ordinary weighted binomial inversion gives

$$H_{a,r} = \sum_{i=0}^{r} (-1)^i \binom{r}{i} f^i T_{a,r-i}. \tag{7}$$

To check this, substitute (1) on the right. The coefficient of $f^{r-s} H_{a,s}$ is

$$\binom{r}{s} \sum_{i=0}^{r-s} (-1)^i \binom{r-s}{i}.$$

This is one if $s = r$ and zero otherwise. No division by f is used. Equation (2) follows because $H_{a,k} = 0$ for $k \geq a$; substitution into (1) gives (3).

Finally the coefficient of $f^{k-j} T_{a,j}$ in (3) is

$$\begin{aligned} \sum_{r=j}^{a-1} (-1)^{r-j} \binom{k}{r} \binom{r}{j} &= \binom{k}{j} \sum_{u=0}^{a-j-1} (-1)^u \binom{k-j}{u} \\ &= (-1)^{a-j-1} \binom{k}{j} \binom{k-j-1}{a-j-1}. \end{aligned}$$

The last equality is the alternating partial-binomial identity (an immediate telescoping consequence of Pascal's rule), valid here because $k \geq a$. This proves (4). □

Corollary 1. *The binomial transforms (7) have nonnegative integer coefficients. Moreover the positional generating function satisfies*

$$g_a(x,t) = \sum_{r=0}^{a-1} \frac{H_{a,r}(x) t^r}{(1 - t f(x))^{r+1}} - T_{a,0}(x).$$

Proof. The first assertion follows from the definition of the cores. For the second, sum (1) over $k \geq 0$ using $\sum_{k \geq r} \binom{k}{r} z^{k-r} = (1-z)^{-r-1}$, and remove the $k = 0$ term. □

Exact corroboration and scope. The accompanying standard-library verifier enumerates permutations through size 9, checks compression and reconstruction for 113,979 objects, and checks 2,808 arbitrary primitive insertions by direct pattern inspection. Two pattern detectors agree on all 46,233 permutations of sizes 1 through 8. It also checks the coefficient identities and their binomial simplification with exact integers. These computations corroborate the proof; the all-parameter conclusion rests on Lemma 1 and the bijection above. The initial core series are not evaluated in closed form here.

Task 2. Explicit initial series (proposed)

Find closed forms or finite functional equations for $H_{3,1}$ and $H_{3,2}$. The corollary would then give an explicit expression for $g_3(x,t)$, addressing the further question stated in the source.

Task 3. Sharpness of the component bound (proposed)

Determine for which a the series $H_{a,a-1}$ is nonzero and find its least nonzero coefficient. This would decide when the degree $a-1$ in the binomial expansion is necessary.

Task 4. Other forbidden patterns (proposed)

Classify permutation classes whose primitive suffix blocks have the same deletion and arbitrary-reinsertion property. Test this first for the extra restrictions 1243 and 2134 studied in [2], and formulate the corresponding finite expansions.

Task 5. Asymptotics at fixed a (proposed)

Determine the dominant singularities of the finitely many $H_{a,r}$, then use the displayed expression for g_a to obtain coefficient asymptotics and the limiting distribution of the distance from a to the maximum. No asymptotic claim is made here.

Status. Task 1 proves the full stated expansion conjecture. Tasks 2–5 are research proposals and carry no additional solution credit. Review and computation were performed by the authoring agent; no independent peer review or proof-assistant certification is claimed.

References

[1] J. B. Gil, O. A. Lopez, and M. D. Weiner, *A positional statistic for 1324-avoiding permutations*, Discrete Mathematics & Theoretical Computer Science 26:1 (2024). doi:10.46298/dmtcs.12629. Current and journal-final text: arXiv:2311.18227v3.

[2] J. B. Gil, O. A. Lopez, and M. D. Weiner, *On separable permutations and three other pairs in the Schröder class*, arXiv:2603.25528v2 (2026).